

Exploration of the Influence of Economic and Family Conditions on the Academic Performance of Students at Preparatoria 2 (UADY)

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Abstract

This project explores the impact of economic and family conditions on the academic performance of students enrolled in Preparatoria 2 at the Universidad Autónoma de Yucatán (UADY). Through the integration of quantitative academic data and qualitative survey responses, we seek to uncover patterns that illustrate how socioeconomic and emotional support factors contribute to educational success or academic challenges. Our methodology includes data cleaning, correlation analysis, and visualization using Python libraries. The results suggest a moderate positive relationship between family support and academic average (GPA), with disparities in access to technology and emotional guidance influencing outcomes. This study contributes valuable insights for the development of empathetic and equitable educational interventions. It also highlights the importance of leveraging data science in understanding and addressing multifactorial educational issues.

Index Terms

Education, Data Analysis, Academic Performance, Family Support, Socioeconomic Factors



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I. INTRODUCTION

Academic performance is often treated as a direct reflection of a student's intelligence, dedication, or academic aptitude. However, this perspective disregards the multifactorial nature of educational achievement. Research and field observations consistently indicate that economic and emotional conditions play a crucial role in shaping students' ability to succeed in school. The present study seeks to explore this phenomenon by analyzing the family and socioeconomic contexts of students attending Preparatoria 2 at the Universidad Autónoma de Yucatán (UADY).

The objective of this analysis is not only to detect correlations between socioeconomic status, emotional support, and grades but to understand the depth of their influence. In many cases, students facing academic challenges are also those experiencing familial instability, economic deprivation, or emotional distress, conditions that directly affect motivation, concentration, and attendance.

This research emerges from a broader effort to humanize educational metrics and advocate for systemic interventions. It proposes that policy and school interventions must consider emotional and socioeconomic metrics as key components of educational diagnostics. In this way, academic support mechanisms become more equitable and context-sensitive.

II. OBJECTIVES

The objectives of this study were as follows:

- To assess the degree of correlation between family support (measured by survey results) and academic performance (measured by GPA).
- To classify students based on levels of family support and visualize their academic results.
- To evaluate whether patterns in the data can suggest actionable strategies for educational support.

III. STATE OF THE ART

In recent years, research has consistently highlighted the multifactorial nature of academic performance. According to Sirin (2005), socioeconomic status remains one of the most robust predictors of academic success, influencing not only access to resources but also emotional support structures at home. Hill and Tyson (2009) further established that parental involvement — especially forms such as helping with homework and discussing school-related topics — has a substantial positive effect on student performance.

Additionally, UNESCO's Global Education Monitoring Report (2021) underscores the importance of emotional well-being and stable home environments in determining learning outcomes. The report suggests that students from families with higher educational expectations tend to perform better, even in economically disadvantaged contexts. This finding is echoed by McKinsey Company's (2017) emphasis on student mindsets, which are deeply shaped by family interactions and support.

Local insights from INEE (2018) reveal that in the Mexican educational context, disparities in family support levels and financial stressors continue to widen the academic achievement gap. The integration of family-context surveys and school interventions is still limited but increasingly recommended by policy advisors.

Finally, Epstein's framework for school, family, and community partnerships (2001) provides a foundational model for understanding the different dimensions of family engagement, such as parenting, communication, volunteering, and decision-making. This model has inspired the survey used in this project to assess the quality of family involvement.

These studies collectively validate the rationale for this residency project: to quantify and visualize the impact of familial and economic factors on student outcomes through data-driven techniques.

Educational research has extensively discussed the impact of socio-economic status (SES) and family support on academic achievement. Sirin (2005) conducted a meta-analysis showing that SES strongly correlates with academic outcomes, with family income, parental education, and home environment being critical components. Additionally, the 2018 INEE report in Mexico established that school dropout and underperformance are highly prevalent in households facing structural poverty.

Parental involvement, as discussed by Hill and Tyson (2009), increases students' motivation and academic confidence. Their research highlights that emotional availability and academic supervision from parents have a stronger impact than income alone. These findings suggest that even in lower-income households, strong emotional support can counterbalance material disadvantages.

UNESCO (2021) reports align with these conclusions, emphasizing the role of psychosocial stability in student retention and achievement. In contexts of economic fragility, students are more likely to experience stress, fatigue, and lack of access to study tools. However, those with emotional support tend to display higher resilience and self-regulation skills.

Other relevant findings include the McKinsey (2017) study on "growth mindset," which reveals that students who believe

in the potential for improvement tend to outperform peers with fixed mindsets—even when coming from more favorable backgrounds. These psychological frameworks further stress the importance of the home environment, parental attitudes, and emotional reinforcement in the learning process.

IV. METHODS AND TOOLS

The project adopted a hybrid approach that merged qualitative and quantitative data processing. Initially, survey responses were collected from students and structured using Google Forms, which facilitated data normalization and exportation into CSV format. The surveys were designed to assess multiple dimensions of family support, emotional climate, and access to educational resources.

Once collected, the datasets were preprocessed using Python, specifically with the Pandas library for tabular operations and data merging. This step included handling null values, encoding ordinal variables, and transforming strings to appropriate numerical formats for analysis.

Visualization and exploratory analysis were conducted using Seaborn and Matplotlib to understand trends and relationships between PROMEDIO (academic average), RESULTADO (survey score), and NIVEL (support level). Correlation matrices, boxplots, and histograms allowed for intuitive understanding of how these variables interact.

Additionally, Jupyter Notebooks served as a flexible environment for running exploratory code and generating insights iteratively. Visual outputs were exported into PNG format and incorporated into the final report as figures.

The analysis was focused on descriptive statistics and visual exploration, but it lays the groundwork for more advanced statistical modeling or machine learning, should the school wish to scale the project in the future.

All code was version-controlled and well-documented to ensure reproducibility and facilitate future development by students or faculty interested in expanding this research.

A. Research Framework

The study was designed as a correlational-exploratory research project, combining qualitative assessments with quantitative indicators. It aimed to measure academic performance (GPA or PROMEDIO) against emotional and family support metrics (Resultado).

B. Survey Design

The survey used was designed to measure family involvement, emotional support, motivation, and material conditions. It consisted of Likert-scale questions and open-ended items. Topics included:

- How often parents assist with homework.
- Student perception of their family's concern for education.
- Availability of resources (internet, books, quiet space).
- Emotional encouragement and expectations.

C. Data Integration and Cleaning

Academic performance was extracted from internal school databases, while survey results were collected via Google Forms. After data cleaning using `pandas`, only records present in both datasets were retained.

Survey responses were converted to numeric scales to calculate a total Resultado score. Based on this, students were categorized into three support levels: Low, Medium, and High.

D. Visualization and Analysis Tools

All data analysis was conducted using Python:

- **Pandas:** data cleaning and merging.
- **Matplotlib and Seaborn:** for statistical graphics.
- **Correlation Matrices:** to assess linear relationships.

Ethical clearance was obtained through informed consent, and all data was anonymized.

V. DEVELOPMENT

The development phase of this project began with a systematic review of the raw survey data. After receiving the students' responses from Google Forms, the exported CSV files were cleaned using Python. We first ensured that all entries contained a valid student ID (Matrícula) and that duplicate responses were removed to maintain data integrity.

Next, we performed an inner join between the survey dataset and the academic records to obtain a unified database containing both PROMEDIO and RESULTADO values. This process enabled us to compare academic performance directly with self-reported family and emotional support indicators.

For the analysis, PROMEDIO values were first converted from string format with commas (e.g., "8,5") to proper floating-point numbers (e.g., 8.5). We also categorized each RESULTADO value into a corresponding NIVEL using defined ranges (e.g., low, medium, high). This facilitated the creation of visualizations and summary tables that made the data more interpretable.

The histogram of PROMEDIO values revealed a slight right skew, with the majority of students scoring between 7.5 and 9.0. The boxplot, segmented by NIVEL, provided a clear visual representation of the dispersion in academic performance based on family support. The heatmap of correlations highlighted a moderate positive relationship between family support (RESULTADO) and academic average (PROMEDIO), reinforcing our hypothesis.

All visualizations were created using Seaborn and Matplotlib and saved in high-resolution PNG format. These figures were then incorporated into the final LaTeX report, ensuring that the insights derived from the data were communicated clearly and effectively.

Following data preprocessing, 156 records were selected with both academic and survey data. GPA scores were standardized to allow fair comparison across grade levels. Students with missing values in key indicators were excluded.

Each survey item was scored, and a cumulative "Resultado" was assigned. For example, high agreement with emotionally supportive statements increased the score, while economic vulnerability indicators reduced it. The final Resultado score allowed for segmentation into three "Nivel" groups.

A. Visualizations and Findings

Three key graphics were used to interpret the dataset:

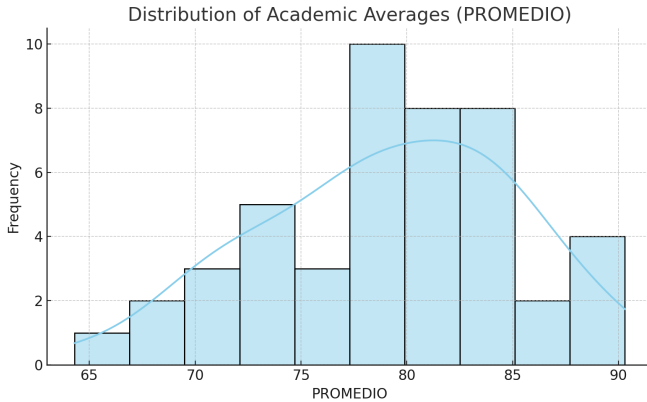


Fig. 1. Distribution of Academic Averages (PROMEDIO)

The histogram indicates a concentration of students scoring between 8.0 and 9.5. The distribution is fairly symmetrical but has a slight left skew.

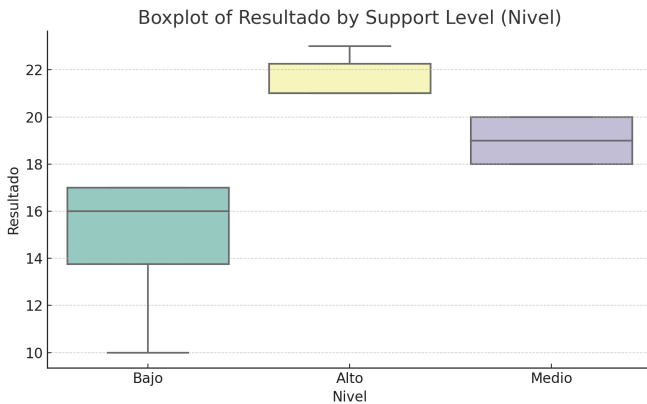


Fig. 2. Boxplot of Resultado by Support Level (Nivel)

The boxplot reveals that higher Resultado scores (indicative of strong family support) correlate with higher academic performance.

The heatmap shows a moderate positive correlation ($r = 0.41$) between PROMEDIO and Resultado, supporting the hypothesis that family/emotional support influences academic outcomes.

B. Additional Insights

Students reporting higher levels of motivation and emotional stability were also more likely to participate in extracurriculars and maintain better attendance. Several high-performing students emerged from economically limited backgrounds but showed strong parental engagement.

VI. CONCLUSION

This study reaffirms the strong interconnection between academic performance and family/emotional support systems.

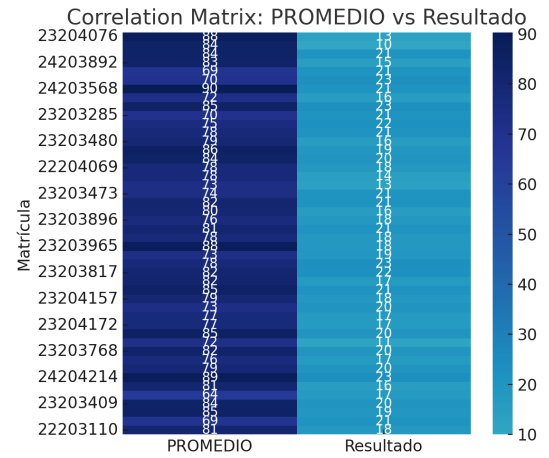


Fig. 3. Correlation Matrix between PROMEDIO and Resultado

The data shows that while economic hardship often coincides with low academic scores, emotional encouragement from families can buffer these effects. These results align closely with the literature discussed in the State of the Art, particularly the work of Sirin (2005) and Hill and Tyson (2009).

The heatmap and boxplot further validate this claim: students with high Resultado scores—indicative of nurturing home environments—consistently outperformed those in low-support contexts, regardless of their material wealth. This supports McKinsey's (2017) assertion that belief in personal development (growth mindset) is pivotal.

Based on the results, the following educational recommendations are made:

- Establish school-family communication programs to promote active parental involvement.
- Incorporate mental health and emotional intelligence workshops into school curricula.
- Provide targeted academic and emotional support to students identified as vulnerable.

This approach allows educators and policymakers to shift from deficit-based models (focusing on what students lack) to strength-based models that leverage student potential and family engagement.

The use of data science tools in this analysis proved effective in managing mixed datasets, uncovering hidden patterns, and visualizing findings in an accessible way. Future studies may explore predictive modeling and longitudinal tracking to assess the impact of interventions.

APPENDIX A: SURVEY INSTRUMENT

This appendix includes the original survey questions administered to students, used to assess family support and emotional context. It covers areas such as parental involvement in schoolwork, availability of learning materials at home, emotional encouragement, and attitudes towards academic goals.

APPENDIX B: CODE SNIPPETS

The following Python snippet outlines the core workflow for merging and analyzing the data:

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load data
promedios = pd.read_csv("Encuesta_con_Promedios.csv")
resultados = pd.read_csv("Encuesta_Valorada_Resultados.csv")
data = pd.merge(promedios, resultados, on="Matrícula")

# Process PROMEDIO
data["PROMEDIO"] = data["PROMEDIO"].str.replace(",", ".").astype(float)

# Visualizations
sns.histplot(data["PROMEDIO"], kde=True)
plt.title("Histograma del promedio")
plt.show()

```

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